

Internship Diary
ON
EIGHT WEEKS OF RESEARCH INTERNSHIP – II

Carried out at
Nitte Centre of Excellence for Applied
Artificial Intelligence

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To



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(Placed under Category 'A' by MHRD, Govt. of India; Accredited as 'A+' Grade University by NAAC)

In partial fulfilment of the requirements for the award of
BACHELOR OF TECHNOLOGY in Information Science and
Engineering

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING



JUNE 2025 – AUGUST 2025

Internship Diary

Day 1 – Monday, 3 June 2025

Focus: Orientation and project overview

Tasks:

- Attended the introductory meeting with my faculty mentor to understand the project objectives related to zero-shot learning and medical imaging.
- Reviewed recent literature on Vision Transformers (ViT) and vision–language models such as CLIP.
- Collected reference papers to understand how zero-shot models are applied in non-medical domains.
- Setup my working environment with access to Google Colab, GitHub repository, and dataset storage.
- Discussed weekly milestones and deliverables with my project partner.

Reflection: Excited to explore transformer models and their relevance to medical image diagnosis.

Day 2 – Tuesday, 4 June 2025

Focus: Understanding dataset and preprocessing requirements

Tasks:

- Downloaded and reviewed the public brain MRI dataset consisting of glioma, meningioma, pituitary, and normal images.
- Verified dataset integrity and ensured consistent image resolutions.
- Studied MRI image formats and learned basic preprocessing operations required for model compatibility.
- Read about common augmentation methods suitable for grayscale MRI images.
- Documented dataset statistics and class distribution in the research log.

Reflection: Realized that handling medical data requires careful normalization and consistency across samples.

Day 3 – Wednesday, 5 June 2025

Focus: Data cleanup and exploratory analysis

Tasks:

- Removed corrupted and duplicated images from the dataset to maintain data quality.

- Visualized sample MRIs for each category to identify underlying variations.
- Applied resizing and RGB conversion pipeline for compatibility with CLIP's image encoder.
- Calculated per-class image counts and balanced the dataset accordingly.
- Saved cleaned dataset versions for zero-shot and few-shot experiments separately.

Reflection: Proper data hygiene builds the foundation for credible experiments.

Day 4 – Thursday, 6 June 2025

Focus: Reviewing CLIP model architecture

Tasks:

- Studied CLIP's dual-encoder design with attention to how images and text are embedded jointly.
- Understood the principle of cosine similarity-based classification.
- Traced how ViT-B/32 processes visual tokens through transformer blocks.
- Noted model input specifications for image dimension and normalization parameters.
- Planned initial experiments to perform zero-shot inference on brain MRIs.

Reflection: Seeing how textual descriptions can replace explicit training labels is fascinating.

Day 5 – Friday, 7 June 2025

Focus: Environment setup for model testing

Tasks:

- Installed necessary Python libraries—including Torch, Transformers, and OpenCLIP.
- Configured GPU runtime in Google Colab for efficient computation.
- Loaded pretrained CLIP weights successfully and validated with sample natural images.
- Verified reproducibility of inference results using random seeds.
- Created base scripts to load medical images and pass them through the CLIP encoder.

Reflection: Achieved a fully working environment—ready to proceed with zero-shot trials.

Day 6 – Saturday, 8 June 2025

Focus: Initial zero-shot experiments

Tasks:

- Constructed simple class prompts such as “an MRI image of a brain with a glioma

tumor.”

- Ran the zero-shot evaluation to test how the pretrained model performs on brain MRIs.
- Observed low accuracy, confirming the expected domain mismatch.
- Logged prediction probabilities and noted confusion patterns.
- Started preparing visual summaries of initial findings.

Reflection: The base model’s struggle emphasizes the uniqueness of medical imagery.

Day 7 – Monday, 9 June 2025

Focus: Error analysis of zero-shot model

Tasks:

- Compared prediction outputs across different prompt phrasings.
- Noted that the model tends to confuse glioma and meningioma classes.
- Checked if normal brain MRI scans are more confidently classified than tumors.
- Documented insights for inclusion in the “Results” section later.
- Prepared a draft of performance metrics and plotted a simple confusion matrix.

Reflection: Pattern analysis helps target which aspects prompt engineering can address.

Day 8 – Tuesday, 10 June 2025

Focus: Research on prompt engineering concepts

Tasks:

- Read scholarly articles about prompt sensitivity in vision–language models.
- Examined how medical terminologies can be incorporated into textual prompts.
- Created an initial prompt set featuring words like “contrast-enhanced,” “abnormal mass,” etc.
- Designed an experimental plan for multi-prompt ensembling.
- Discussed the approach with my mentor for feasibility.

Reflection: Crafting the right linguistic context might bring semantic alignment improvements.

Day 9 – Wednesday, 11 June 2025

Focus: Implementation of domain-specific prompts

Tasks:

- Programmed scripts to encode multiple prompts per class.

- Implemented averaging of text embeddings for class-level vectors.
- Ran evaluations using new prompts and recorded performance changes.
- Compared results versus the generic prompt baseline.
- Visualized similarity heatmaps between image and text embeddings.

Reflection: Noticed modest improvement—language context does make a measurable difference.

Day 10 – Thursday, 12 June 2025

Focus: Analyzing prompt-engineering outcomes

Tasks:

- Computed per-class precision and recall to identify consistent gains.
- Found pituitary and normal classes benefited more from rewritten descriptions.
- Experimented with alternative prompt templates including anatomical location cues.
- Consolidated best-performing prompt variations for later use.
- Wrote observations for the “Prompt Engineering Strategy” section draft.

Reflection: Prompt diversity helps but cannot substitute for domain adaptation entirely.

Day 11 – Friday, 13 June 2025

Focus: Planning few-shot experiments

Tasks:

- Selected 20 representative samples per class for few-shot training.
- Split remaining data into validation and test subsets.
- Outlined training protocols for frozen and partially unfrozen setups.
- Defined metrics and stopping criteria to prevent overfitting.
- Ensured reproducible data splits saved to disk.

Reflection: Careful dataset partitioning is crucial to a fair comparison.

Day 12 – Saturday, 14 June 2025

Focus: Building the frozen-backbone pipeline

Tasks:

- Extracted embeddings from the frozen CLIP vision encoder.
- Implemented a linear classifier head on top of fixed embeddings.
- Normalized features and used balanced cross-entropy loss for training.
- Tuned learning rate and batch size suitable for small datasets.

- Monitored training curve stability over several epochs.

Reflection: Linear probing provides insights into the inherent transferability of features.

Day 13 – Monday, 16 June 2025

Focus: Evaluating frozen-backbone results

Tasks:

- Computed accuracy, F1-score, and confusion matrix for the trained model.
- Compared performance against zero-shot baselines.
- Noted minimal improvement—around 2% absolute gain.
- Analyzed embeddings visually using PCA projections.
- Discussed strategies to enable better domain adaptation.

Reflection: Feature alignment remains poor without at least partial fine-tuning.

Day 14 – Tuesday, 17 June 2025

Focus: Studying fine-tuning strategies

Tasks:

- Read literature on partial unfreezing and parameter-efficient adaptation.
- Identified which transformer blocks contain higher-level semantic information.
- Planned to unfreeze the final transformer block in the vision encoder.
- Designed a mixed-learning rate schedule for frozen and trainable layers.
- Added gradient clipping to stabilize training.

Reflection: Selective fine-tuning seems promising for limited-data regimes.

Day 15 – Wednesday, 18 June 2025

Focus: Implementation of partial unfreezing

Tasks:

- Modified training script to unfreeze the last transformer block.
- Verified that trainable parameters reduced to a small fraction of the total.
- Initiated training with small batch size and low learning rate.
- Logged training losses to ensure no divergence.
- Saved checkpoints across epochs for later evaluation.

Reflection: Controlled adaptation maintains stability yet introduces necessary flexibility.

Day 16 – Thursday, 19 June 2025

Focus: Monitoring adaptation training

Tasks:

- Continued training for several epochs, tracking validation accuracy.
- Observed smoother convergence compared to the fully frozen setup.
- Fine-tuned the classifier head jointly with the unfrozen block.
- Conducted mini-experiments varying learning rates.
- Recorded results for later aggregation.

Reflection: Partial fine-tuning begins to yield noticeably better generalization.

Day 17 – Friday, 20 June 2025

Focus: Evaluation of partially fine-tuned model

Tasks:

- Ran inference on the fixed test set and computed all evaluation metrics.
- Achieved a significant gain—around 60% accuracy overall.
- Generated updated confusion matrix highlighting improved separation.
- Compared per-class metrics with zero-shot and frozen baselines.
- Drafted a short experimental note summarizing performance gains.

Reflection: Minimal adaptation indeed bridges a large portion of the domain gap.

Day 18 – Saturday, 21 June 2025

Focus: Comparative analysis

Tasks:

- Consolidated all results (zero-shot, prompt-engineered, few-shot, partial).
- Created summary tables and charts for clear comparison.
- Verified metric calculations and resolved minor computation discrepancies.
- Drafted early form of Table I for the paper.
- Discussed interpretations with co-authors.

Reflection: Quantitative comparisons strongly justify the proposed approach.

Day 19 – Monday, 23 June 2025

Focus: Visual result exploration

Tasks:

- Generated embeddings plotted with t-SNE to visualize class clustering.
- Overlayed partial fine-tuned embeddings showing improved separability.

- Analyzed false positives to identify pattern overlaps.
- Selected representative MRI samples illustrating model predictions.
- Stored plots for inclusion in the “Results and Discussion” section.

Reflection: Visualization provides intuitive confirmation of numerical improvements.

Day 20 – Tuesday, 24 June 2025

Focus: Addressing overfitting concerns

Tasks:

- Re-ran training with slightly stronger data augmentations.
- Tested different optimizer hyperparameters for stability.
- Evaluated model on unseen validation subsets.
- Confirmed consistent results within ± 2 % variance.
- Documented exact configuration details for reproducibility.

Reflection: Proper validation ensures credibility of reported improvements.

Day 21 – Wednesday, 25 June 2025

Focus: Drafting methodology section

Tasks:

- Wrote comprehensive explanation of zero-shot framework and partial adaptation.
- Illustrated processing pipeline from image preprocessing to similarity computation.
- Added details about prompt construction and averaging mechanism.
- Ensured mathematical notation for similarity computation is clear.
- Shared draft with mentor for editing suggestions.

Reflection: Writing down the pipeline consolidates conceptual understanding.

Day 22 – Thursday, 26 June 2025

Focus: Dataset description and problem formulation

Tasks:

- Composed dataset paragraph describing MRI acquisition variability.
- Highlighted challenges like grayscale intensity and subtle boundaries.
- Defined problem as a four-class classification task.
- Described data splits and few-shot training configuration.
- Included preprocessing diagram placeholder for the paper figure.

Reflection: Clear articulation of problem setup strengthens research transparency.

Day 23 – Friday, 27 June 2025

Focus: Writing related-work section

Tasks:

- Summarized CNN-based methods and their limitations.
- Added subsections on transformers in medical imaging and zero-shot learning.
- Highlighted recent works such as ViT, CLIP, and UNETR.
- Made sure citations correspond to the correct references.
- Drafted connecting paragraphs ensuring logical flow.

Reflection: Reviewing prior work positions our contribution in a broader context.

Day 24 – Saturday, 28 June 2025

Focus: Structuring introduction

Tasks:

- Prepared background explaining the significance of MRI analysis in oncology.
- Explained challenges of low-resource scenarios.
- Added motivation for exploring vision–language models.
- Wrote concise statements for problem definition and contributions.
- Revised introduction flow with mentor’s feedback.

Reflection: Strong introductions help readers grasp the research direction quickly.

Day 25 – Monday, 30 June 2025

Focus: Implementation details section

Tasks:

- Recorded all training hyperparameters, dataset preprocessing, and augmentation details.
- Noted usage of AdamW optimizer and learning-rate settings for reproducibility.
- Added technical specifications of computing environment (Google Colab GPU).
- Structured subsection headings aligned with IEEE formatting.
- Created Figure 1 pipeline image caption draft.

Reflection: Precision in reporting experiments enhances scientific reliability.

Day 26 – Tuesday, 1 July 2025

Focus: Drafting quantitative results section

Tasks:

- Inserted results table showing accuracy improvements across methods.

- Explained the interpretation of observed trends.
- Drafted subheadings for quantitative and qualitative analysis.
- Linked metrics to clinical relevance, e.g., improved recall for critical classes.
- Proofread numeric consistency throughout.

Reflection: Translating data into coherent narrative showcases impact effectively.

Day 27 – Wednesday, 2 July 2025

Focus: Constructing confusion-matrix visualizations

Tasks:

- Generated confusion matrices for all evaluated settings.
- Color-coded plots for clarity in grayscale print format.
- Integrated them into manuscript figure placeholders.
- Analyzed which tumor patterns were misclassified frequently.
- Noted points for discussion on model limitations.

Reflection: Visual comparisons make the improvement story compelling.

Day 28 – Thursday, 3 July 2025

Focus: Drafting discussion section (part 1)

Tasks:

- Interpreted why zero-shot inference fails in medical domains.
- Connected failure reasons to domain shift and training data bias.
- Discussed prompt engineering's partial success.
- Described overfitting tendencies in frozen few-shot models.
- Structured subtopics to flow toward the success of partial adaptation.

Reflection: Discussion writing deepens my grasp of experimental implications.

Day 29 – Friday, 4 July 2025

Focus: Drafting discussion section (part 2)

Tasks:

- Explained how partial unfreezing balances generalization and adaptation.
- Added notes on interpretability and clinical applicability.
- Compared findings with related studies for context.
- Identified open issues like glioma-meningioma overlap.
- Prepared closing paragraph linking results to future enhancements.

Reflection: Summarizing outcomes provides closure to experimental reasoning.

Day 30 – Saturday, 5 July 2025

Focus: Writing conclusion section

Tasks:

- Summarized key findings succinctly.
- Reinforced central claim on the effectiveness of limited fine-tuning.
- Suggested future directions such as multimodal integration and adapters.
- Drafted take-home message for real-world applicability.
- Ensured alignment between abstract and conclusion messages.

Reflection: Crafting a concise conclusion amplifies the overall impact of the study.

Day 31 – Monday, 7 July 2025

Focus: Reference formatting and citation checking

Tasks:

- Cross-verified all in-text citations with the reference list.
- Converted entries to IEEE reference style.
- Checked consistency in authors' names and publication details.
- Added missing DOIs and conference information.
- Ensured alphabetical ordering and sequential numbering in the manuscript.

Reflection: Proper citations uphold academic integrity.

Day 32 – Tuesday, 8 July 2025

Focus: Figure refinement

Tasks:

- Re-designed framework diagram for clarity.
- Adjusted label fonts and arrows for professional look.
- Created schematic of the training pipeline using vector graphics.
- Saved all figures in high-resolution formats for insertion.
- Updated figure captions according to IEEE standards.

Reflection: Well-styled visuals convey technical ideas effectively.

Day 33 – Wednesday, 9 July 2025

Focus: Mentor review session

Tasks:

- Presented compiled draft to mentor and research team.
- Received feedback on flow and clarity.

- Noted requests for shortening certain repetitive descriptions.
- Discussed potential additions to emphasize novelty.
- Scheduled subsequent corrections based on recommendations.

Reflection: Constructive feedback brings out clarity and focus in the manuscript.

Day 34 – Thursday, 10 July 2025

Focus: Implementing feedback revisions

Tasks:

- Simplified dense paragraphs in methodology.
- Added explanation for hyperparameter choices.
- Clarified figure references inside text.
- Fixed minor grammatical inconsistencies pointed out by the mentor.
- Ran plagiarism check to verify originality percentage.

Reflection: Early revisions strengthen coherence before final editing rounds.

Day 35 – Friday, 11 July 2025

Focus: Enhancing readability and flow

Tasks:

- Rewrote sentences for smoother transitions between sections.
- Combined overlapping ideas in “Discussion” and “Conclusion.”
- Checked for consistent tense and voice throughout the paper.
- Adjusted paragraph spacing and layout.
- Updated numbering of tables and figures.

Reflection: Continuous polishing transforms a technical draft into a readable paper.

Day 36 – Saturday, 12 July 2025

Focus: Validation experiments re-run

Tasks:

- Repeated selected experiments to confirm reproducibility.
- Verified accuracy metrics remained consistent with earlier runs.
- Exported model checkpoints for archival.
- Generated updated log files and compared parameter counts.
- Stored all experiment configuration JSONs for future use.

Reflection: Verification adds confidence before paper submission.

Day 37 – Monday, 14 July 2025

Focus: Ethical considerations and dataset documentation

Tasks:

- Drafted a subsection describing data source and ethics compliance.
- Clarified that dataset used is publicly available and de-identified.
- Wrote lines about privacy and regulatory limitations of medical data.
- Inserted acknowledgment of dataset creators.
- Proofread ethics statement for accuracy.

Reflection: Transparency about data sources is essential for responsible AI research.

Day 38 – Tuesday, 15 July 2025

Focus: Abstract rewriting

Tasks:

- Condensed initial abstract for precision and readability.
- Highlighted key results and comparative improvements.
- Rephrased sentences to emphasize motivation, methods, and contributions.
- Ensured alignment with the main body.
- Checked total word count fits within guidelines.

Reflection: A concise abstract effectively summarizes months of effort.

Day 39 – Wednesday, 16 July 2025

Focus: Formatting according to conference template

Tasks:

- Applied IEEE two-column layout styles.
- Adjusted margin, font sizes, and figure placements.
- Ensured all headings follow hierarchies I–II–III consistently.
- Included keywords, author affiliations, and corresponding author email formatting.
- Verified PDF compatibility across devices.

Reflection: Proper formatting gives the manuscript a professional appearance.

Day 40 – Thursday, 17 July 2025

Focus: Proofreading technical accuracy

Tasks:

- Checked all numerical results, ensuring correct reporting in text and tables.
- Cross-validated confusion-matrix values with computed metrics.

- Eliminated redundant repetition of results in multiple sections.
- Re-ran grammar and spell-check tools.
- Highlighted areas requiring final mentor verification.

Reflection: Accurate data presentation preserves credibility.

Day 41 – Friday, 18 July 2025

Focus: Peer review among teammates

Tasks:

- Exchanged drafts with team members for informal peer review.
- Collected comments on structure and clarity.
- Discussed suggestions over a short call.
- Integrated minor stylistic improvements noted by peers.
- Appreciated collaborative editing insights.

Reflection: Peer collaboration adds diversity in perspective.

Day 42 – Saturday, 19 July 2025

Focus: Visualization polish and caption editing

Tasks:

- Adjusted color palettes of graphs for better readability.
- Added detailed captions explaining findings.
- Ensured figure numbering matches references within text.
- Uploaded images to the LaTeX source folder.
- Generated final high-resolution PDF for inspection.

Reflection: Clear visuals strengthen overall comprehension for readers.

Day 43 – Monday, 21 July 2025

Focus: Mentor feedback round 2

Tasks:

- Received secondary review comments from mentor and co-author.
- Discussed the need to streamline the methodology section wording.
- Added clarifying notes on evaluation metrics and cross-institution reproducibility.
- Reconfirmed internal consistency of results.
- Scheduled two days for final proofreading round.

Reflection: Iterative revisions ensure the manuscript communicates ideas effectively.

Day 44 – Tuesday, 22 July 2025

Focus: Implementing second-round corrections

Tasks:

- Simplified jargon in certain technical sentences.
- Reduced paragraph length in “Discussion.”
- Enhanced explanations for readers with limited machine-learning background.
- Corrected reference cross-links and numbering.
- Verified PDF rendering of formulae and symbols.

Reflection: Accessibility of technical writing widens the audience reach.

Day 45 – Wednesday, 23 July 2025

Focus: Table and result consistency checks

Tasks:

- Ensured all tables are captioned and labeled consistently.
- Verified numerical entries against raw CSV logs.
- Checked percentage rounding errors.
- Updated legends for better interpretability.
- Finalized Table I summarizing accuracy comparisons.

Reflection: Data traceability adds robustness to scientific reporting.

Day 46 – Thursday, 24 July 2025

Focus: Grammar and language refinement

Tasks:

- Read the paper aloud to identify awkward phrasing.
- Adjusted sentence flow for smoother narration.
- Replaced repetitive connectors with varied transitions.
- Ensured subject–verb agreement throughout.
- Re-checked spacing, punctuation, and capitalization.

Reflection: Linguistic refinement substantially improves readability.

Day 47 – Friday, 25 July 2025

Focus: Final image placement and aesthetic layout

Tasks:

- Positioned figures optimally within columns to improve visual balance.
- Verified images are near their first textual mention.

- Adjusted captions and figure references accordingly.
- Recompiled LaTeX multiple times until no format warnings appeared.
- Compared print preview for visual alignment.

Reflection: Attention to presentation details gives the paper a polished finish.

Day 48 – Saturday, 26 July 2025

Focus: Comprehensive paper read-through

Tasks:

- Conducted beginning-to-end reading of manuscript for coherence.
- Checked title clarity, abstract alignment, and conclusion emphasis.
- Ensured consistent terminology (e.g., “few-shot” vs. “low-shot”).
- Verified no missing figure or table references remain unresolved.
- Compiled list of final-minute observations.

Reflection: A holistic review reveals subtle inconsistencies often missed earlier.

Day 49 – Monday, 28 July 2025

Focus: Mentor approval and polishing

Tasks:

- Sent near-final version for official mentor review.
- Received approval with minor comments on abstract phrasing.
- Applied requested refinements and corrected typographical slips.
- Checked authors’ affiliations one last time.
- Exported updated PDF to shared drive.

Reflection: The project nears completion with most aspects finalized.

Day 50 – Tuesday, 29 July 2025

Focus: Final quality assurance

Tasks:

- Conducted plagiarism check ensuring originality threshold under acceptable limits.
- Verified figure resolutions meet publication requirements.
- Checked reference hyperlinks in PDF.
- Reviewed formatting consistency between sections.
- Created final backup copies of all assets.

Reflection: Diligent verification ensures readiness for submission.

Day 51 – Wednesday, 30 July 2025

Focus: Abstract and keyword optimization

Tasks:

- Refined keyword list emphasizing core concepts: zero-shot learning, transformers, low-resource diagnosis.
- Shortened abstract sentences for better readability.
- Replaced redundant terms with concise alternatives.
- Reviewed alignment of keywords with journal indexing standards.
- Updated metadata fields in PDF.

Reflection: Small language improvements increase paper discoverability.

Day 52 – Thursday, 31 July 2025

Focus: Preparing presentation slides

Tasks:

- Created slide deck summarizing the research workflow.
- Included data visualization and result comparison charts.
- Practiced explaining methodology succinctly.
- Added diagrams of partial transformer unfreezing for clarity.
- Coordinated with teammates for upcoming presentation schedule.

Reflection: Presenting work visually helps identify remaining clarity gaps.

Day 53 – Friday, 1 August 2025

Focus: Rehearsal and performance analysis

Tasks:

- Delivered mock presentation to colleagues.
- Collected suggestions for pacing and slide transitions.
- Modified slides to highlight key contributions.
- Improved articulation of results significance to non-technical audience.
- Recorded presentation timing for smooth flow.

Reflection: Effective communication translates technical details into impactful insight.

Day 54 – Saturday, 2 August 2025

Focus: Preparing paper submission package

Tasks:

- Compiled all required files—manuscript, figures, tables, and references.
- Verified that margins and fonts comply with submission portal checks.
- Filled metadata forms including abstract, author list, and affiliations.
- Uploaded documents to online submission system for internal review.
- Cross-checked everything against checklist before final upload tomorrow.

Reflection: Attention to submission details avoids unnecessary last-minute issues.

Day 55 – Sunday, 3 August 2025

Focus: Final review and submission

Tasks:

- Performed last spell-check and read through entire PDF.
- Ensured figures and tables retain correct labels after conversion.
- Received final consent from mentor for official submission.
- Submitted the paper to the targeted conference system successfully.
- Noted submission confirmation and stored receipt copy.

Reflection: Submitting the paper feels deeply satisfying after weeks of focused effort.

Day 56 – Monday, 4 August 2025

Focus: Reflection and closure

Tasks:

- Compiled entire research logbook, experiment notebooks, and results summary.
- Wrote a reflection essay on technical learning outcomes during the internship.
- Discussed potential extensions like multimodal fusion with my mentor.
- Cleaned up the project repository and archived final models.
- Prepared internship completion report including this daily diary.

Reflection: The internship has been an enriching journey—learning both research discipline and practical implementation of advanced AI in medical imaging.